

# Video Game Sales

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**ABSTRACT** The project examines the factors that influence video game global sales using the Video Game Sales with Ratings dataset. Its main objective is to develop an accurate predictive model for Global\_Sales by analyzing game attributes such as platform, genre, publisher, ratings, and regional sales. The methodology includes thorough data preprocessing to handle missing values, standardize data types, and clean categorical variables. A hybrid feature selection approach combining Pearson Correlation and Random Forest Feature Importance is applied to identify the most relevant predictors. For modeling, the study employs Support Vector Regression (SVR) to capture both linear and nonlinear relationships within the data. This approach enables a deeper understanding of sales-driving factors and demonstrates the effectiveness of data science techniques in forecasting real-world market performance.

## I. INTRODUCTION

The video game industry has experienced significant growth over the last twenty years, establishing itself as one of the most lucrative entertainment sectors globally. It is crucial for developers, publishers, and marketing teams to comprehend the elements that affect a game's commercial success to create products that resonate with player preferences and adapt to changing market dynamics. As game-related information becomes increasingly accessible, data science methodologies offer robust tools for deriving insights and facilitating data-informed decision-making. In this project, we tackle the challenge of forecasting **Global\_Sales** of video games by utilizing historical data from the publicly available dataset "Video Game Sales with Ratings." This dataset encompasses various attributes including platform, genre, release year, publisher, regional and global sales, critic ratings, user ratings, and ESRB classifications. However, despite its comprehensive nature, the dataset presents challenges such as missing values, inconsistent formats, and a multitude of categorical variables. These complications necessitate thorough preprocessing to guarantee precise predictions and insightful analysis.

The primary aim of this project is to cultivate a data-driven comprehension of the factors influencing video game sales and to develop a predictive model that can estimate global sales for upcoming titles. By employing data science techniques such as data cleansing, exploratory analysis, encoding of categorical features, and dataset preparation for modeling, we seek to pinpoint the characteristics that most significantly impact a game's success. This endeavor underscores the importance of leveraging data science

approaches on real-world industry data, illustrating how unrefined information can be converted into valuable insights for gaming companies, developers, and marketing professionals.

## II. LITERATURE REVIEW

Recent research conducted between 2021 and 2025 has focused on predicting video game sales through the application of statistical analysis and machine learning methodologies. These studies have pinpointed essential factors that affect global sales and have underscored the difficulties associated with modeling such data.

**Study 1** performed a descriptive analysis of long-term trends in the video game market [1]. It recognized genre, publisher, and regional markets as significant contributors to sales performance. However, it did not account for trends emerging after 2020, which diminishes its applicability to current predictive tasks.

**Study 2** employed econometric models including correlation, regression, and logit/probit methods to evaluate platform, genre, and regional sales as indicators of global performance [2]. While it validated these variables as important, issues such as multicollinearity and inconsistencies in the data compromised the reliability of the model, indicating a necessity for more robust modeling approaches.

**Study 3** conducted a comparative analysis of machine learning models for forecasting global video game sales, which included Linear Regression, Random Forest, KNN, Gradient Boosting, and SVM [3]. Following preprocessing and evaluation via MAE, SVM demonstrated superior performance, suggesting its capability in managing nonlinear

relationships. The occurrence of overfitting in tree-based models emphasized the requirement for more rigorous validation techniques.

**Study 4** presented a hybrid feature selection strategy that integrated correlation analysis with the importance derived from Random Forest [4]. This methodology enhanced model accuracy and identified platform, genre, and publisher as critical predictors. Nonetheless, the study was limited by the absence of cross-dataset testing and hyperparameter optimization, which restricted its generalizability.

**Study 5** implemented a hybrid optimization strategy utilizing Elastic Net and MOPSO to improve feature selection and minimize prediction error [5]. Although effective, this method proved to be computationally demanding and was evaluated on a limited number of datasets.

The studies that were reviewed indicated that Support Vector Machine (SVM) outperformed other machine learning models in terms of prediction accuracy [3]. Given that our research focuses on forecasting continuous values, specifically **global\_sales**, we will employ Support Vector Regression (SVR), which serves as the regression counterpart to SVM and is well-suited for this type of analysis. Furthermore, Study 4 emphasized that the application of feature selection techniques enhances the accuracy of models [4]. Consequently, our methodology will integrate SVR with a suitable feature selection approach to attain more dependable and precise predictions.

### III. DATASETS AND METHODOLOGY

#### A. DATA COLLECTION, DESCRIPTION, AND PREPROCESSING

The dataset was sourced from Kaggle [6], a well-known platform for data science datasets and competitions. It was selected because it provides comprehensive, real-world sales information across multiple platforms, genres, and regions, making it directly relevant to the research question: "What factors most strongly predict a video game's global sales?". This combination allows for an analysis of the relationship between a game's critical reception, popularity and its commercial success, making it ideal for investigating market trends and publishing strategies.

The dataset (*Video Game Sales with Ratings*) contains over 16,000 video game entries, where each row represents a unique game. Its structure includes three main types of information:

- 1) **Identifying Information:** Name, Platform, Genre, Publisher, Developer, and Year\_of\_Release.
- 2) **Sales Data:** NA\_Sales, EU\_Sales, JP\_Sales, Other\_Sales, and Global\_Sales (in millions).

- 3) **Ratings and Reception:** Critic\_Score (0–100), User\_Score (0–10), Critic\_Count, and User\_Count.

However, the dataset includes notable limitations. Several key fields, such as Year\_of\_Release, Critic\_Score, User\_Score, and ESRB Rating contain missing values. The dataset also tends to overrepresent high-selling commercial titles and underrepresent indie or low-selling games. User\_Score values may be biased due to potential manipulation such as review bombing. Additionally, the dataset ends around 2016, meaning it does not reflect recent gaming trends such as digital distribution, subscription services, or live-service models.

#### B. DATA SCIENCE TECHNIQUES OVERVIEW

The dataset underwent several preprocessing steps to ensure clean, consistent, and model-ready data. First, non-useful columns such as Name, Developer, Rating, and User\_Score were removed because they contained too many missing values, inconsistent formatting, or provided no predictive value. Next, data types were corrected: the Year\_of\_Release column was converted to integer format, while critic/user score and count fields were converted to numeric.

Missing values were then handled using a simple but effective strategy, critic/user review metrics were filled with 0 (indicating no reviews), while categorical fields such as Publisher and Genre were filled with "Unknown". Afterward, categorical variables underwent encoding:

- Platform: Frequency Encoding
- Genre: One-Hot Encoding
- Publisher: Target Encoding

Finally, because Global\_Sales is highly skewed, a new variable (Log\_Global\_Sales) was created using a log transformation to stabilize variance and improve model performance. All preprocessing steps were executed in Python using pandas, NumPy, and matplotlib, preparing the cleaned dataset for model implementation.

#### C. METHODOLOGY

This research employs a systematic approach that comprises two primary elements: feature selection and predictive modeling. The aim is to determine the key factors that influence **Global\_Sales** and to create a dependable regression model that can effectively represent both linear and nonlinear relationships present in the data.

##### 1) Feature Selection

A hybrid feature selection approach was employed to reduce noise, improve model performance, and retain only the most relevant variables. This approach integrates two complementary techniques:

**Pearson Correlation Coefficient (PCC):** PCC was first applied to measure the strength of the linear relationship between each feature and the target variable. Features exhibiting weak or negligible correlations were removed.

**Random Forest Feature Selection (RFFS):** The remaining features were then evaluated using a Random Forest model to determine their relative importance. Unlike PCC, which captures only linear relationships, RFFS identifies nonlinear patterns and evaluates how strongly each feature contributes to prediction accuracy.

By combining statistical filtering through PCC with model-based ranking through RFFS, the hybrid method ensures that only meaningful and influential features are retained.

## 2) Predictive Modeling Using Support Vector Regression

Support Vector Regression (SVR) is designed to predict continuous values, such as the **global\_sales** of a video game, by creating a model that balances accuracy with the ability to generalize well to new data. Unlike traditional regression, which tries to reduce the overall error as much as possible, SVR focuses on keeping the prediction function as simple and smooth as possible. It introduces an  $\epsilon$ -insensitive margin, which is a small zone around the predicted function where errors are ignored. Any data point that falls within this margin does not affect the model. Only points with errors greater than  $\epsilon$  influence the training process. These important points are called support vectors, and they directly determine the final shape of the regression function. To control how strictly the model fits the data, SVR uses a regularization parameter  $C$ . A higher value of  $C$  tries to fit more points exactly, while a lower value allows more flexibility and helps avoid overfitting. This makes SVR stable and reliable, especially when working with real-world datasets that may contain noise or inconsistencies. Because many relationships in video game sales are nonlinear for example, how genre, platform, and ratings interact. SVR uses kernel functions to model these patterns. The Radial Basis Function (RBF) kernel is commonly used because it transforms the original features into a higher-dimensional space, allowing the model to capture curved or complex trends that a simple linear model cannot [7].

## IV. PRELIMINARY ANALYSIS

### A. INITIAL STEPS

A detailed preprocessing procedure was carried out to ensure the dataset was suitable for analysis. Several attributes contained missing or inconsistent values. Therefore, handling them properly was a critical first step.

Based on the dataset "Video Game Sales with Ratings" which is coming with attributes such as title, platform, year of release, genre, publisher, regional sales, global sales, critic and user ratings, developer, and ESRB rating. We are working hard to understand these columns and what can help in our goal which is predicting the `global_sales` and we end up with the following points for data cleaning and preparation.

#### 1) Handling Missing and Inconsistent Values

The preprocessing phase began with identifying missing values across key columns. Several fields related to critic and user reviews contained missing entries, while Publisher and Genre also included incomplete values. Review-related missing values were filled with 0 (representing no reviews), and categorical fields were filled with "Unknown" to maintain dataset consistency. Columns irrelevant to prediction such as Name, Developer, Rating, and User\_Score were removed due to high missingness or low predictive usefulness.

#### 2) Data Type Conversion

Numerical fields were standardized to appropriate formats. Year\_of\_Release was converted to integer, and critic/user score and count fields were coerced into numeric types to ensure proper processing. All sales columns remained numerical since they represent sales volumes.

#### 3) Encoding Categorical Features

Three categorical variables required encoding for modeling. Platform was frequently encoded due to its large number of unique categories. Genre was transformed using one-hot encoding. Publisher was encoded using target encoding based on mean sales, which helps capture publisher-level patterns while avoiding excessive dimensionality.

#### 4) Log Transformation of the Target Variable

Because Global\_Sales is highly skewed, a log-transformed version (`Log_Global_Sales`) was created using  $\log(1 + x)$ . This transformation reduces skewness and improves the performance of regression models.

## B. DATA EXPLORATION

### 1) Descriptive Statistics

The initial exploration analysis began with a review of descriptive statistics to characterize the structure and distribution of the dataset. The `Global_Sales` variable exhibited a highly right-skewed distribution, with most video games selling fewer than one million units, while a small number of blockbuster titles achieved exceptionally high sales. This extreme skewness motivated the later decision to

apply a log transformation during preprocessing to stabilize variance for predictive modeling. Examination of temporal patterns revealed that the number of game releases increased steadily throughout the 1990s and early 2000s, peaking around 2008–2009 before declining in subsequent years. Regional sales summaries showed that North America and Europe dominate total sales, while Japan and other regions contribute significantly less, reflecting well-known differences in regional gaming markets.

Analysis of categorical variables highlighted notable imbalances. Action games represent the most common genre, whereas Puzzle and Strategy games are comparatively rare. Platform distributions indicated that the PS2 and Nintendo DS have the highest number of released titles. Interestingly, older Nintendo platforms such as the NES, SNES, and Game Boy despite having fewer titles exhibited particularly high average sales due to the presence of historically iconic best-selling games. A missing-value assessment revealed that critic and user review metrics are absent primarily in older games, a pattern consistent with the fact that many early titles were released before modern review aggregation systems existed. This indicates that the missingness is not random but structurally related to release year.

## 2) Hypothesis Formulation

Based on observations from descriptive analysis, several research hypotheses were formulated to investigate potential relationships between game characteristics and commercial performance. Each hypothesis is presented with its corresponding null ( $H_0$ ) and alternative ( $H_1$ ) statements.

### Hypothesis 1: Effect of Critic Score on Sales

Because visual inspection suggested a positive association between critic ratings and global sales, we tested whether higher critic scores correspond to significantly higher sales.

**$H_0$ :** There is no difference in mean global sales between games with high critic scores and those with low critic scores.

**$H_1$ :** Games with higher critic scores have significantly higher mean global sales.

### Hypothesis 2: Sales Differences Between Action and Non-Action Genres

Action games are highly represented in the dataset, but descriptive observations did not clearly indicate whether they sell better than other genres.

**$H_0$ :** Action games and non-Action games have equal mean global sales.

**$H_1$ :** Action games and non-Action games differ in mean global sales.

### Hypothesis 3: Review Count Differences Before and After 2005

Because critic-review systems became more widespread after the mid-2000s, we tested whether review counts increased significantly for newer titles.

**$H_0$ :** There is no difference in critic review counts between games released before 2005 and after 2005.

**$H_1$ :** Games released after 2005 have significantly higher critic review counts.

### Hypothesis 4: Sales Performance of Nintendo vs. Other Publishers

Descriptive analysis showed that Nintendo titles frequently dominate top-selling lists, motivating a publisher-level comparison.

**$H_0$ :** Nintendo and non-Nintendo publishers have equal mean global sales.

**$H_1$ :** Nintendo-published games have higher mean global sales than games published by other companies.

## 3) Inferential Statistical Techniques

To evaluate these hypotheses, inferential statistical methods including two-sample t-tests and z-score analysis, were applied to the dataset.

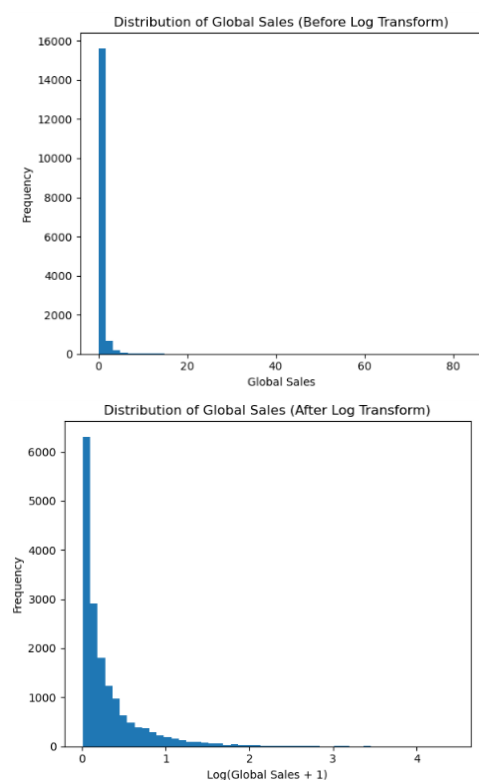
A t-test comparing games above and below the median critic score revealed a highly significant difference in global sales ( $p = 1.34 \times 10^{-73}$ ), confirming that higher critic ratings are strongly associated with higher commercial performance. This provides robust statistical support for the hypothesis that critical reception positively influences sales outcomes. Next, a t-test comparing Action games to all other genres returned a non-significant result ( $p = 0.419$ ). Despite Action being the most common genre, this finding indicates that Action titles do not differ meaningfully in average global sales from other genres. Thus, genre frequency does not necessarily correspond to higher sales performance.

A third t-test assessed whether critic review counts differed before and after 2005. The test returned to a non-numeric (NaN) p-value, caused by the presence of substantial missing data among older games. This outcome is informative, as it validates the earlier observation that review availability is structurally dependent on release year. Older games frequently lack critic review information altogether, making formal comparison impossible without extensive imputation. A further comparison between Nintendo and non-Nintendo publishers yielded a highly significant difference in global sales ( $p = 1.51 \times 10^{-21}$ ), confirming that Nintendo titles

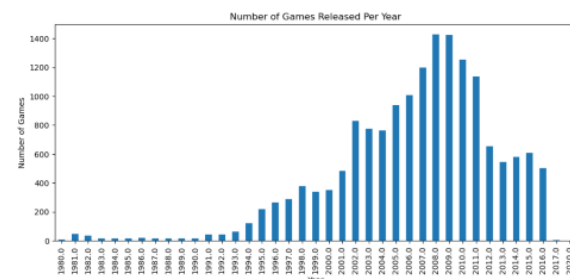
achieve substantially higher average sales. This statistically supports the descriptive observation that Nintendo produces fewer games overall but consistently releases top-selling titles.

### C. VISUALIZATIONS AND FINDINGS

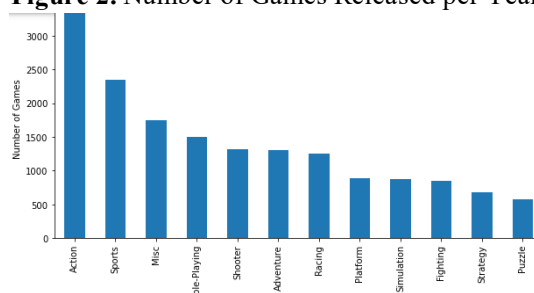
The visual analysis revealed several clear trends in the dataset. The distribution of Global\_Sales showed a strongly right-skewed pattern, indicating that while most games achieve relatively low sales, a small number become extremely high sellers. A yearly release trend plot demonstrated rapid industry growth through the 1990s and 2000s, peaking around 2008–2009 before declining, which also helps explain gaps in review data for older titles. Bar charts for genre and platform frequencies highlighted that Action games dominate the market in quantity, while platforms such as PS2 and DS host the largest number of releases. However, average-sales visualizations showed a different perspective, revealing that Platform and Shooter games generally achieve higher performance, and several older Nintendo platforms exhibit unusually high average sales due to iconic best-selling titles. Together, these graphics provide meaningful insights that guided both the understanding of industry behavior and the modeling strategy.



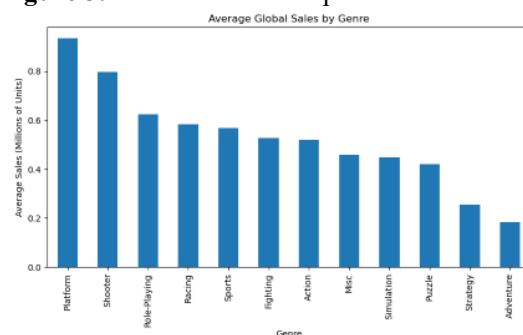
**Figure 1.** Distribution of Global Sales (before and after log transform)



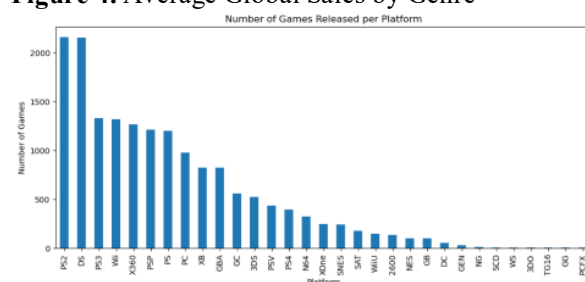
**Figure 2.** Number of Games Released per Year



**Figure 3.** Number of Games per Genre



**Figure 4.** Average Global Sales by Genre



**Figure 5.** Number of Games per Platform



**Figure 6.** Average Global Sales by Platform



## V. FINAL ANALYSIS AND MODEL BUILDING

### A. REFINEMENT OF ANALYSIS

Following the exploratory analysis, the dataset was refined to prepare it for modeling. Because Global\_Sales showed extreme right-skew, a log transformation was applied to improve model stability. Missing critic and user review values common in older games were set to zero rather than removed. Non-informative or inconsistent features such as Name, Developer, Rating, and User\_Score were dropped due to high missingness or weak predictive value. Categorical variables were encoded using different strategies based on their characteristics. Platform (31 categories) was encoded with Frequency Encoding to avoid the dimensional explosion of one-hot encoding. Genre (12 categories) use One-Hot Encoding because differences between genres are meaningful and manageable in size. Publisher (581 categories) used Target Encoding, which captures average sales performance without generating hundreds of sparse dummy variables.

### B. MODEL BUILDING

The refined dataset was split into training and testing sets, and numerical features were standardized using **StandardScaler**. Feature selection combined Pearson correlation for linear relationships and Random Forest importance scores for nonlinear relevance, producing a focused and effective feature set.

Support Vector Regression (SVR) with an RBF kernel was selected as the primary model due to its ability to handle nonlinear patterns and scaled data. After preprocessing and feature selection, SVR achieved an  $R^2$  score of approximately 0.83, indicating strong predictive performance.

### C. ADDITIONAL EXPERIMENTS

Different SVR settings were tested to improve model performance.

## VI. RESULTS

The Support Vector Regression (SVR) model performed strongly after the refined preprocessing and feature-selection steps. Using the log-transformed global sales as the target, the final SVR model achieved an  $R^2$  score of approximately 0.83 on the test set, indicating that it explained most of the variability in game sales.

## VII. CONCLUSION

This project investigates the key factors influencing global video game sales using data preprocessing, exploratory analysis, feature selection, and machine learning. The results

show that regional sales especially NA\_Sales and EU\_Sales have the strongest relationship with Global\_Sales, while platform and genre also contribute significantly when modeled correctly. The study further demonstrates that combining multiple feature selection techniques improves prediction accuracy, particularly with datasets containing missing or imbalanced values. Although the dataset provides useful insights into the gaming industry, it is limited by its time range and uneven representation of certain genres and companies. Overall, the findings highlight the importance of data analysis in understanding market dynamics and guiding decisions in game development and marketing. Future work could enhance results by expanding data sources or applying more advanced models such as optimized SVR or hybrid techniques.

## VIII. INDIVIDUAL CONTRIBUTIONS

The successful completion of this project reflects the collective effort of all group members, with each individual contributing to specific components of the work. Fatimah Alabdellah prepared the abstract, while Norah Almari developed the introduction section. Atekah Aljaafar authored the conclusion, and Fatimah Albeshir completed the literature review. In addition, all members collaborated on the data preprocessing, methodology development, feature selection, and analytical processes, ensuring coherence, accuracy, and alignment across all sections of the project.

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